RCS05 | Student Packet

CTS Cohort 6 | CTS Day 13 | 2026-07-27 | 10:00 AM-3:00 PM | 6 students

Advanced Electrical: Switch, Light, and Troubleshooting

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Troubleshooting is disciplined evidence gathering: verify the source state, follow the path, find the first failed checkpoint, correct one issue, and retest.

Learning targets

- Trace a de-energized switch-controlled lighting path
- Install a training switch and light outlet box
- Use visual inspection and continuity to diagnose inserted faults
- Document the fault, evidence, correction, and retest

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and energy-boundary check

10:50-11:20 | Switch-controlled lighting path model

11:20-12:00 | Splice, pigtail, and termination demonstration

12:00-12:30 | Troubleshooting ladder guided practice and lab release

12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Switch/light install and inserted-fault lab

2:25-2:45 | Correction, retest, evidence, cleanup

2:45-3:00 | Exit ticket and Day 6 preview

Key words

line | load | switch leg | open circuit | short circuit | fault | continuity | splice | pigtail | verification

Guided worksheet

1. Draw the switch-controlled light path.

_____

_____

2. Predict three faults that could create an open circuit.

_____

_____

3. Write the safe continuity-test sequence.

_____

_____

4. Record expected and observed results at each checkpoint.

_____

_____

5. Explain why changing two things at once weakens the diagnosis.

_____

_____

Afternoon lab mission

Students add one training switch box and one light outlet box to the capstone wall, complete de-energized terminations, then diagnose an instructor-inserted open-circuit or misconnection using visual and continuity evidence.

Safety boundary

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Instructor verifies isolation before testing
- No student energized diagnostics
- Continuity only on isolated conductors
- Replace damaged devices or conductors
- Close and label the training system after approval

Build / test sequence

[ ] Energy-control check

[ ] Review diagram

[ ] Lay out boxes

[ ] Mount and route

[ ] Prepare splice/pigtails

[ ] Terminate switch and light path

[ ] Instructor inspection

[ ] Receive faulted state

[ ] Predict and test checkpoints

[ ] Correct one fault

[ ] Retest

[ ] Document and reset

Evidence record

[ ] Lighting-path diagram: _____

[ ] Switch/light installation QC: _____

[ ] Troubleshooting log: _____

[ ] Before/after test results: _____

[ ] Instructor verification: _____

Reflection and exit ticket

Draw line-switch-load-neutral

_____

_____

Name the first failed checkpoint

_____

_____

State the energy boundary

_____

_____

Describe the evidence that proved your correction

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____